



USER MANUAL
BMW M2 CS



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DEAR iRACING USER,

Congratulations on your purchase of the BMW M2 CSR! From all of us at iRacing, we appreciate your support and your commitment to our product. We aim to deliver the ultimate sim racing experience, and we hope that you'll find plenty of excitement with us behind the wheel of your new car!

The following guide explains how to get the most out of your new car, from how to adjust its settings off of the track to what you'll see inside of the cockpit while driving. We hope that you'll find it useful in getting up to speed.

Thanks again for your purchase, and we'll see you on the track!





CHASSIS

**STRUT FRONT SUSPENSION,
MULTI-LINK REAR SUSPENSION**



LENGTH
4460 mm
175.6 in

WIDTH
1854 mm
73 in

WHEELBASE
2692 mm
106 in

DRY WEIGHT
1544 kg
3405 lbs

WET WEIGHT
WITH DRIVER
1715 kg
3781 lbs

POWER UNIT

SIX-CYLINDER IN-LINE, FOUR VALVES



DISPLACEMENT
3.0 Liters
183 cid

TORQUE
406 lb-ft
550 Nm

POWER
365 bhp
272 kW

RPM LIMIT
7500



INTRODUCTION

The information found in this guide is intended to provide a deeper understanding of the chassis setup adjustments available in the garage, so that you may use the garage to tune the chassis setup to your preference.

Before diving into chassis adjustments, though, it is best to become familiar with the car and track. To that end, we have provided baseline setups for each track commonly raced by these cars.

To access the baseline setups, simply open the Garage, click iRacing Setups, and select the appropriate setup for your track of choice. If you are driving a track for which a dedicated baseline setup is not included, you may select a setup for a similar track to use as your

baseline. After you have selected an appropriate setup, get on track and focus on making smooth and consistent laps, identifying the proper racing line and experiencing tire wear and handling trends over a number of laps.

Once you are confident that you are nearing your driving potential with the included baseline setups, read on to begin tuning the car to your handling preferences.

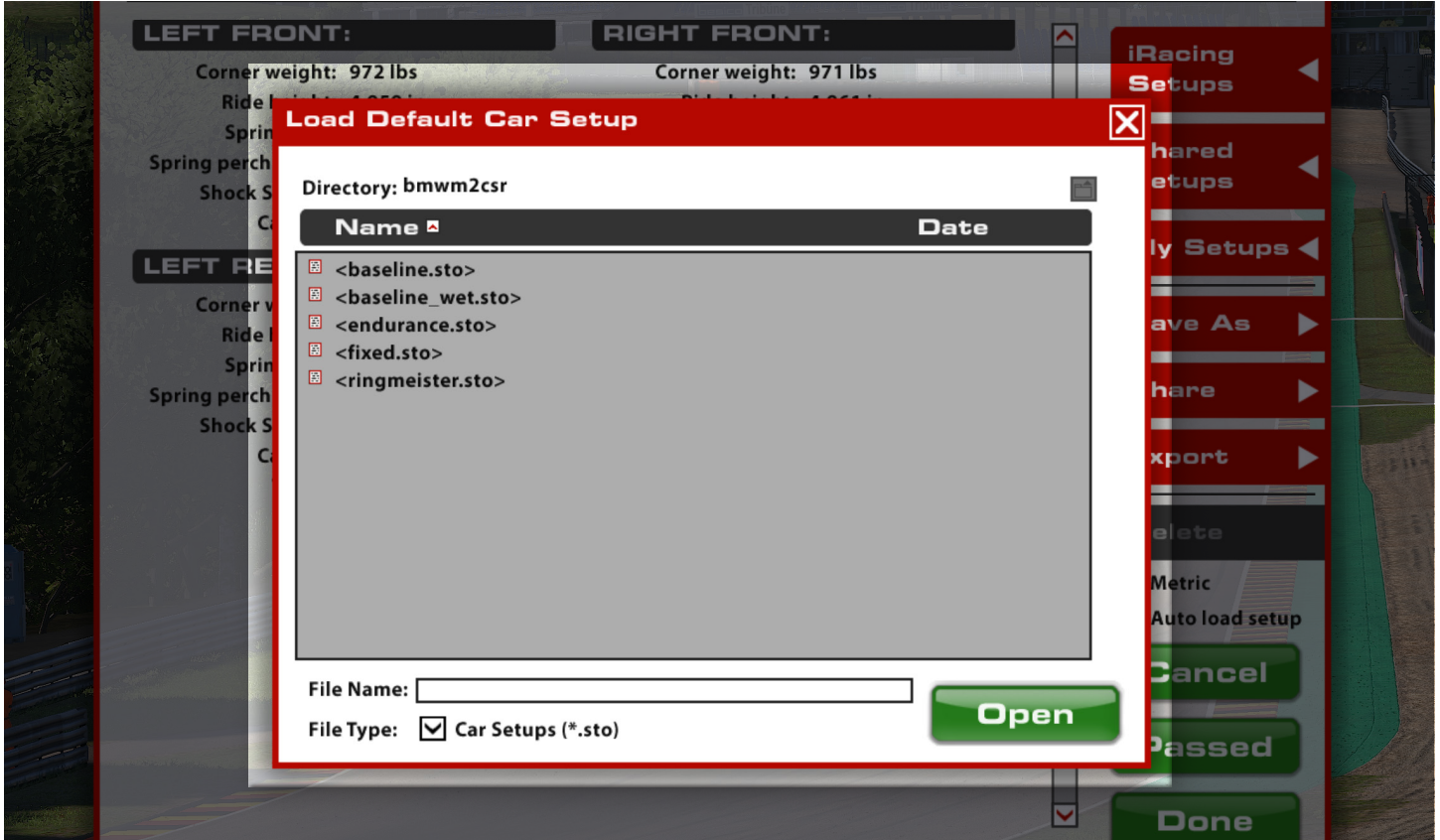
GETTING STARTED



Once you load into the car, getting started is as easy as selecting the “upshift” button to put it into gear, and hitting the accelerator pedal. This car uses a sequential transmission and does not require a clutch input to shift in either direction. However the car’s downshift

protection will not allow you to downshift if it feels you are traveling too fast for the gear selected and would incur engine damage. If that is the case, the gear change command will simply be ignored.

LOADING AN iRACING SETUP



Upon loading into a session, the car will automatically load the iRacing Baseline setup <baseline.sto>. If you would prefer one of iRacing's pre-built setups that suit various conditions, you may load it by clicking Garage > iRacing Setups > and then selecting the setup to suit your needs.

If you would like to customize the setup, simply make the changes in the garage that you would like to update and click apply. If you would like to save your setup for future use click "Save As" on the right to name and save the changes. To access all of your personally saved setups, click "My Setups" on the right side of the garage.

If you would like to share a setup with another driver or everyone in a session, you can select "Share" on the right side of the garage to do so.

If a driver is trying to share a setup with you, you will find it under "Shared Setups" on the right side of the garage as well.

DASH PAGES

A single-page dash display provides all relevant information to the driver, as well as status LEDs and a cluster of shift lights above the display.



Tyre Pressure	The live tire pressure readouts are shown in the upper left of the display
Oil	Engine Oil temperature, in °C or °F
WAT	Engine cooling water temperature, in °C or °F
Tire Temperatures	Live tire carcass temperatures for each tire, in °C or °F. These values mirror what is seen on the Garage's Tire page.
TDiffOil	Rear differential oil temperature, in °C or °F
GBX	Gearbox oil temperature, in °C or °F
Lap Number	Number of laps completed in the current session
Best Time	The current session's best lap time
Predictive Time	Displays the estimated lap time for the current lap based on live splits
nEng	Engine RPM
Speed	Current vehicle speed, in kph or mph
GFuelUsed	Amount of fuel used since leaving the pits

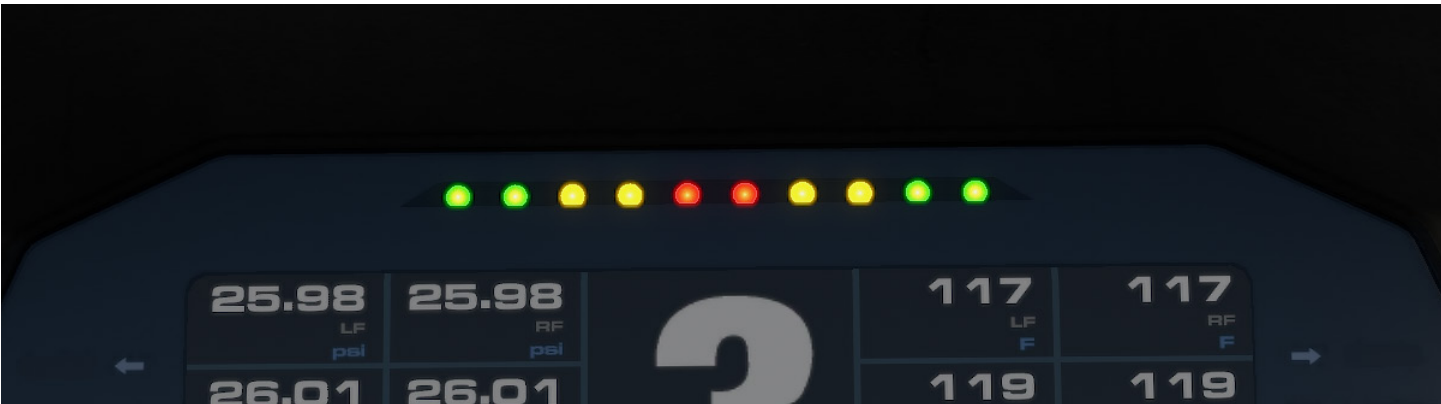
STATUS LIGHTS

Eight lights are arranged on either side of the digital dash to display when various systems are active in the car.



Fuel	Illuminates in red when the amount of fuel in the car drops below 9 liters (2.4 gallons)
PSL	Flashes yellow when the Pit Speed Limiter is active
DSC	Changes color based on the current DSC setting: Red when the system is OFF, Blue when the system is in MDM mode, and Green when the system is ON.
Lights	Illuminates when the headlights and accessory exterior lights are on
ABS	Inoperable, illuminates in the event of an ABS system failure
FDS	Changes color to represent the current FDS (Throttle) setting: Blue for “Dry” mode, Green for “Wet” mode

SHIFT LIGHTS

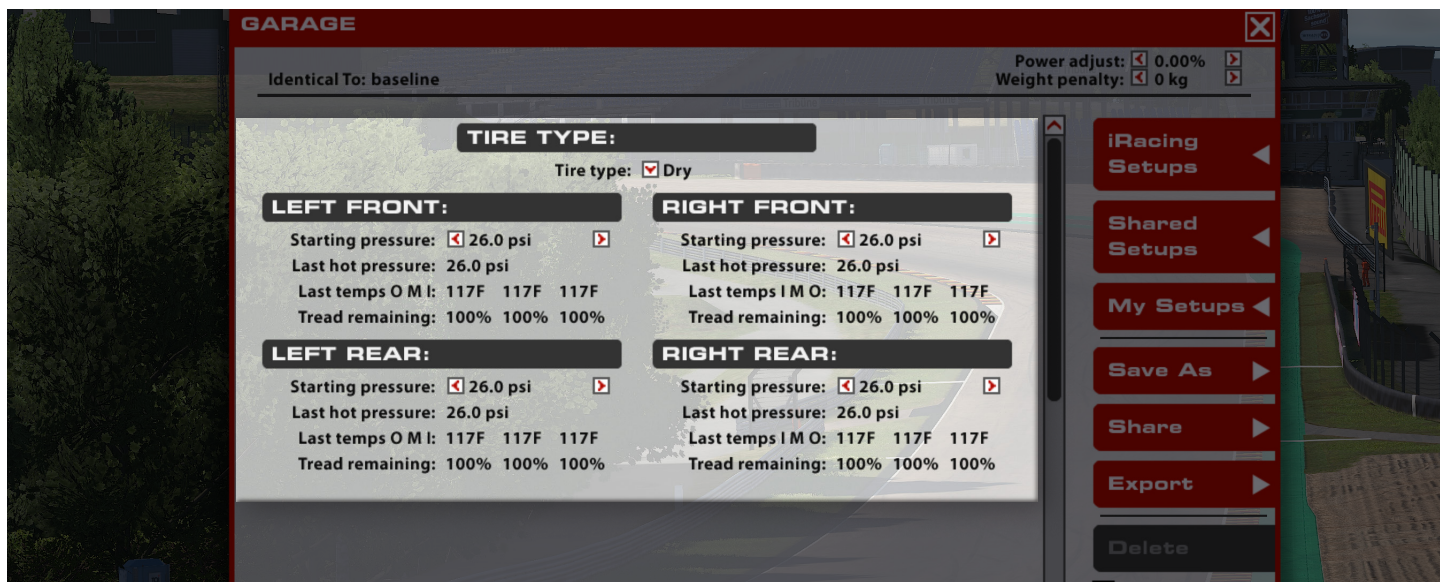


A row of ten LED lights sit above the digital display to show when the engine RPM has reached the optimal shift point. Upshifting should be done when all the lights are illuminated.

ADVANCED SETUP OPTIONS

This section is aimed toward more advanced users who want to dive deeper into the different aspects of the vehicle's setup. Making adjustments to the following parameters is not required and can lead to significant changes in the way a vehicle handles. It is recommended that any adjustments are made in an incremental fashion and only singular variables are adjusted before testing changes.

TIRES



TIRE TYPE

Selects which type of tire is installed on the car when loaded into the world. Dry, or slick, tires are used for dry racing conditions while Wet tires are intended for raining and wet track conditions.

COLD PRESSURE / STARTING PRESSURE

The air pressure in the tires when the car is loaded into the world. Lower pressures will provide more grip but will produce more rolling drag and build temperature faster. Higher pressures will feel slightly more responsive and produce less rolling drag, but will result in less grip. Generally, higher pressures are preferred at tracks where speeds are higher while lower pressures work better at slower tracks where mechanical grip is important.

LAST HOT PRESSURE

When the car returns to the garage after an on-track stint, the tire pressure will be displayed as Hot Pressure. The difference between cold and hot pressure is a good way to see how tires are being loaded and worked while on track. Tires seeing more work will build more pressure, and paying attention to which tires are building more pressure and adjusting cold pressure to compensate can be crucial for optimizing tire performance.

LAST TEMPERATURES

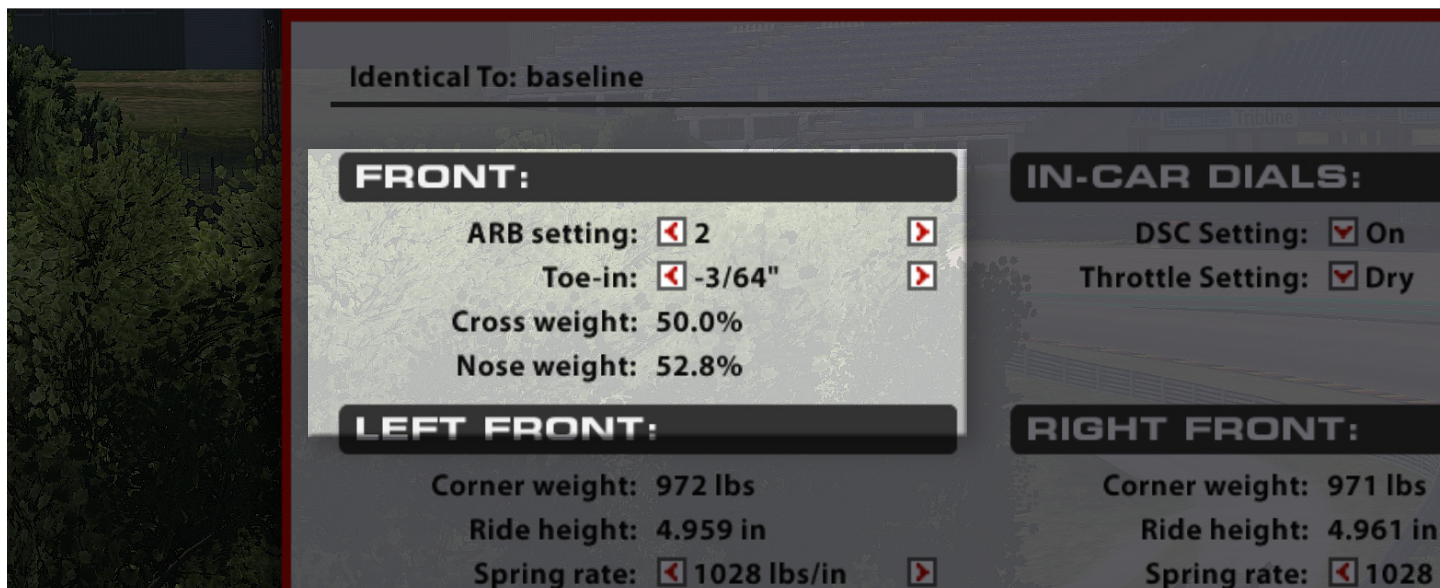
The tire carcass temperatures (measured within the tread) are displayed after the car returns from the track. These temperatures are an effective way to determine how much work or load a given tire is experiencing while on track. Differences between the inner and outer temperatures can be used to tune individual wheel alignment and the center temperatures can be compared to the outer temperatures to help tune tire pressure.

TREAD REMAINING

The amount of tread on the tire, displayed as a percentage of a new tire, is shown below the tire temperatures. These values are good for determining how far a set of tires can go before needing to be replaced, but don't necessarily indicate an under- or over-worked tire in the same way temperatures will.

CHASSIS

FRONT



ARB SETTING

The front ARB setting alters the anti-roll bar stiffness in the front suspension. Two numerical settings are available. Setting 1 producing less front roll stiffness, reducing understeer, and setting 2 producing more front roll stiffness, increasing understeer.

TOE-IN

Toe is the angle of the wheel, when viewed from above, relative to the centerline of the chassis. Toe-in is when the front of the wheel is closer to the centerline than the rear of the wheel, and Toe-out is the opposite. On the front end, adding toe-out (negative value) will increase slip in the inside tire and decrease straight-line stability while adding toe-in will reduce the slip and increase straight-line stability.

CROSS WEIGHT

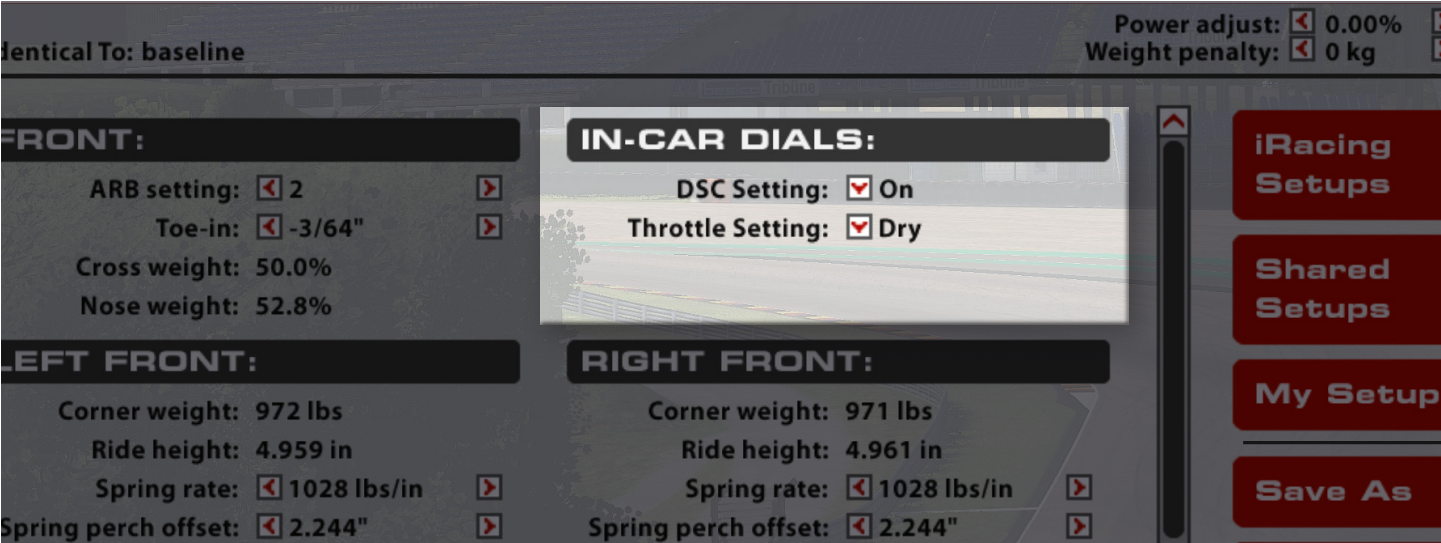
The percentage of total vehicle weight in the garage acting across

the right front and left rear corners. A setting of 50.0% is generally optimal for non-oval tracks as this will produce symmetrical handling in both left and right hand corners providing all other chassis settings are symmetrical. Higher than 50% cross weight will result in more understeer in left hand corners and increased oversteer in right hand corners. Cross weight can be adjusted by making changes to the spring perch offsets at each corner of the car.

NOSE WEIGHT

The percentage of total vehicle weight in the garage across the front axle. Higher Nose Weight values will induce understeer when cornering and promote straight-line stability, while lower Nose Weight values will increase oversteer while cornering. This is not directly adjustable and is altered by other adjustments, such as Fuel Level.

IN-CAR DIALS



DSC SETTING

Allows adjustment of the dynamic stability control (DSC) system. Three options are available, 'ON' - Full assist with aggressive torque cut when wheelspin is detected, 'MDM' - Reduced intervention and 'OFF'. Typically, MDM and OFF are used. MDM provides a good balance between assistance in large slides or moments of excessive wheelspin while allowing some degree of slip in normal usage. It is recommended to learn the car with DSC set to MDM before transitioning to OFF.

THROTTLE SETTING

The throttle setting alters how much power and torque is delivered through the throttle pedal's range of movement. The Dry option will deliver more power at low throttle values, while the Wet option will reduce power delivery at low throttle positions to help prevent wheelspin in low-grip conditions or in the rain.



FRONT CORNERS



CORNER WEIGHT

The weight underneath each tire under static conditions in the garage. Correct weight arrangement around the car is crucial for optimizing a car for a given track and conditions. Individual wheel weight adjustments and crossweight adjustments are made via the spring perch offset adjustments at each corner.

RIDE HEIGHT

Distance from the ground to the bottom surface of the car's floor on the front wheel centerline. Adjusting Ride Heights is key for optimum performance, as they can directly influence the vehicle's aerodynamic performance as well as mechanical grip. Increasing front ride height will decrease front downforce as well as decrease overall downforce, but will allow for more weight transfer across the front axle when cornering. Conversely, reducing ride height will increase front and overall downforce, but reduce the weight transfer across the front axle.

SPRING RATE

This setting determines the installed corner spring stiffness. Stiffer springs will result in a smaller variance in ride height between high and low load cases and will produce superior aerodynamic performance through improved platform control. However overly stiff springs will result in increased tire load variation which will manifest as a loss in mechanical grip. Typically the drawbacks of stiffer springs will become more pronounced on rougher tracks and softer springs in these situations will result in increased overall performance. Corner spring changes will influence both roll and pitch control of the platform and ARB changes should be considered when altering corner spring stiffnesses in order to retain the same front to rear roll stiffness and overall balance. When reducing corner spring stiffness the ARB stiffness should be increased to retain the same roll stiffness as previously. Three options are available and they adjust symmetrically (changing the left-front will change the right-front, and vice versa).

SPRING PERCH OFFSET

Used to adjust the ride height at the corner of the car by changing the installed position of the spring. Increasing the spring perch offset will result in lowering the corner of the car while reducing the spring perch offset will raise the corner of the car. These changes should be kept symmetrical across the axle (left to right) to ensure the same corner ride heights and no change in cross weight. The spring perch offsets can also be used in diagonal pairs (LF to RR and RF to LR) to change the static cross weight in the car.

SHOCK SETTING

The shock setting is a paired adjustment, changing both the compression and rebound side of the shock through a single setting. Settings are represented with a negative value, with "-1" being the stiffest setting and "-20" being the softest setting. Stiffer shocks will control the chassis's attitude and ride heights better while sacrificing bump absorption, while softer shocks will absorb bumps better at the cost of chassis height control.

CAMBER

Camber is the vertical angle of the wheel relative to the center of the chassis. Negative camber is when the top of the wheel is closer to the chassis centerline than the bottom of the wheel, positive camber is when the top of the tire is farther out than the bottom. Due to suspension geometry and corner loads, negative camber is desired on all four wheels. Higher negative camber values will increase the cornering force generated by the tire, but will reduce the amount of longitudinal grip the tire will have under braking. Excessive camber values can produce very high cornering forces but will also significantly reduce tire life, so it is important to find a balance between life and performance. Increasing front camber values will typically result in increased front axle grip during mid to high speed cornering but will result in a loss of braking performance and necessitate a rearward shift in brake bias to compensate.

REAR CORNERS



CORNER WEIGHT

The weight underneath each tire under static conditions in the garage. Correct weight arrangement around the car is crucial for optimizing a car for a given track and conditions. Individual wheel weight adjustments and crossweight adjustments are made via the spring perch offset adjustments at each corner.

RIDE HEIGHT

Distance from ground to a reference point on the rear of the chassis. Increasing rear ride height will decrease rear downforce as well as increase overall downforce and will allow for more weight transfer across the rear axle when cornering. Conversely, reducing ride height will increase rear downforce percentage but reduce overall downforce while reducing the weight transfer across the rear axle. Rear ride height is a critical tuning component for both mechanical and aerodynamic balance considerations and static rear ride heights should be considered and matched to the chosen rear corner springs for optimal performance.

SPRING RATE

Similar to the front axle, stiffer springs will result in a smaller variance in ride height between high and low load cases and will produce superior aerodynamic performance through improved platform control at the expense of mechanical grip. This can be particularly prominent when exiting slow speed corners with aggressive throttle application. Stiffer springs will tend to react poorly during these instances especially so on rough tracks which will result in significant traction loss. Spring stiffness should be matched to the needs of the racetrack and set such that the handling balance is consistent between high and low speed cornering. As an example case, a car which suffers from high speed understeer but low speed oversteer could benefit from an increase in rear spring stiffness. This will allow for a lower static rear height which will reduce rear weight transfer during slow speed cornering while maintaining or even increasing the rear ride height in high speed cornering to shift the aerodynamic balance forwards and reduce understeer. Three options are available and they adjust symmetrically (changing the left-front will change the right-front, and vice versa).

SPRING PERCH OFFSET

Used to adjust the ride height at the corner of the car by changing the installed position of the spring. Increasing the spring perch offset will result in lowering the corner of the car while reducing the spring perch offset will raise the corner of the car. These changes should be kept symmetrical across the axle (left to right) to ensure the same corner ride heights and no change in cross weight. The spring perch offsets can also be used in diagonal pairs (LF to RR and RF to LR) to change the static cross weight in the car.

SHOCK SETTING

The shock setting is a paired adjustment, changing both the compression and rebound side of the shock through a single setting. Settings are represented with a negative value, with "-1" being the stiffest setting and "-20" being the softest setting. Stiffer shocks will control the chassis's attitude and ride heights better while sacrificing bump absorption, while softer shocks will absorb bumps better at the cost of chassis height control.

CAMBER

As with the front of the car it is desirable to run significant amounts of negative camber in order to increase the lateral grip capability; however, it is typical to run slightly reduced rear camber relative to the front. This is primarily for two reasons, firstly, the rear tires are wider compared to the fronts and secondly the rear tires must also perform the duty of driving the car forwards where benefits of camber to lateral grip become a tradeoff against reduced longitudinal (traction) performance.

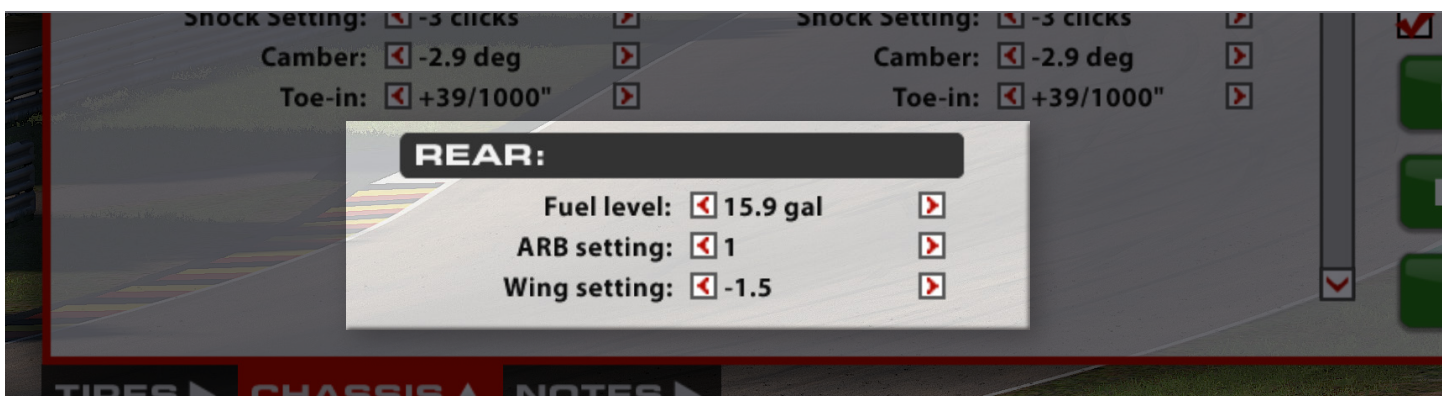


TOE-IN

Toe is the angle of the wheel, when viewed from above, relative to the centerline of the chassis. Toe-in is when the front of the wheel is closer to the centerline than the rear of the wheel, and Toe-out is the opposite. At the rear of the car it is typical to run toe-in. Increases in toe-in will result in improved straight line stability and a reduction in response during direction changes. Large values of toe-in should be avoided if possible as this will increase rolling drag and reduce straight line speeds. When making rear toe changes remember that

the values are for each individual wheel as opposed to paired as at the front. This means that individual values on the rear wheels are twice as powerful as the combined adjustment at the front of the car when the rear toes are summed together. Generally, it is advised to keep the left and right toe values equal to prevent crabbing or asymmetric handling behavior; however, heavily asymmetric tracks such as Lime Rock Park may see a benefit in performance from running asymmetric configurations of rear toe and other setup parameters.

REAR



FUEL LEVEL

The amount of fuel in the fuel tank when the car is loaded into the world.

ARB SETTING

Like with the front suspension, the rear suspension's Anti-Roll Bar can be adjusted via the rear ARB setting. Two options are available, with setting 1 producing less rear roll stiffness, inducing understeer, and setting 2 producing more rear roll stiffness, promoting oversteer.

WING SETTING

The wing setting refers to the relative angle of attack of the rear wing, this is an aerodynamic device which has a significant impact upon the total downforce (and drag) produced by the car as well as shifting the aerodynamic balance of the car rearwards with increasing angle. Increasing the rear wing angle results in more total cornering grip capability in medium to high speed corners but will also result in a reduction of straight line speed. Rear wing angle should be adjusted in conjunction with front and rear ride heights, specifically the difference between front and rear ride heights known as 'rake'. To retain the same overall aerodynamic balance it is necessary to increase the rake of the car when increasing the rear wing angle.

SETUP TIPS

This section is aimed toward helping users who want to dive deeper into the different aspects of the vehicle's setup.

SETUP TIPS

PROVIDED SETUPS

If the setup fails tech inspection, it is likely the ride heights require adjustment. This is performed by using the spring perch adjustments at either end of the car. Right clicks (positive) will decrease the ride height while left clicks (negative) will increase the ride height.

In the iRacing Setups folder you will find a variety of setups.

Baseline is a 50% fuel load setup which is intended solely for loading the car. As such, this setup should always pass tech inspection at every fuel load and track but will not provide ultimate performance.

SETUP NOTES

The BMW M2 CSR is a car with relatively few setup adjustments available, however, should you wish to build your own setup, we have provided the following notes to help you get started.

TIRE PRESSURES

For all 4 corners at most tracks minimum air pressures will likely provide the most performance, these minimums have been set such that the hot pressures reach a typical real-world hot pressure target. However, should the track have particularly long straightaways (such as Le Mans) then you may benefit from increasing tire pressures slightly to reduce rolling drag and tire temperature generation.

WING SETTING

Two positions of wing angle are available, a higher wing angle produces more aerodynamic downforce, more drag and a rearwards shift in aero balance. For tracks with long straights it is beneficial to run at the -4 position while tracks with many high speed corners (or hot conditions) may benefit from the extra stability provided by the -1.5 position.

SPRING RATE

At the front of the car three options of spring rate are available while at the rear five are available. Both front and rear have similar total authority to change the mechanical balance but with the rear having a finer range of adjustment available. At the front of the car a stiffer spring will result in more understeer but reduced aero pitch sensitivity while at the rear of the car a stiffer spring will result in more oversteer.

CAMBER

At the front of the car it is advised to run as much camber as possible. More camber results in more lateral (cornering) capability while reducing longitudinal (traction and straight line braking) capability. Often the gains in lateral capability will outweigh the loss in longitudinal capability however, you may find the trade off of running lower camber at the rear of the car (such as approximately -2.0 degrees) to be worthwhile in respect to corner exit traction.

Setups labeled '_wet' have wet tyres pre-fitted and setup adjustments to suit wet conditions.

Setups labeled '_endurance' have a 100% fuel load and are for use where no fuel restriction is present and/or race lengths are approximately 1 hour or more in length.

The setup titled 'fixed' is the setup used in the fixed setup series

ARB SETTING

Two positions of ARB are available at each end of the car as a fine tune to the balance or to offset a change in corner spring rate. At the front of the car a stiffer ARB will result in more understeer while at the rear of the car a stiffer ARB will result in more oversteer. A change in front ARB is approximately equal to one position change in front spring rate, while at the rear of the car a change in rear ARB is approximately equal to two position changes in rear spring rate. For example, If the front spring is changed from 180 N/mm to 200 N/mm then a corresponding change in the front ARB from 2 → 1 will restore the mechanical balance approximately to its prior setting.

RIDE HEIGHT

At both the front and rear of the car the most performance mechanically and aerodynamically is achieved at the lowest possible ride heights while still passing tech inspection. This is 125 mm in the front and rear. Front ride height should only be increased at extreme tracks such as the Nurburgring in the event of excessive ground contact. Rear ride height can be increased slightly to move the aerodynamic balance forwards for more oversteer but this will produce less overall downforce, more aerodynamic drag and reduced rear axle capability unless it is paired with an increase in rear wing angle. This is adjusted by moving the spring perch offsets as stated previously.

TOE-IN

At the front of the car more toe-in can result in improved mid corner performance but at the cost of entry stability and straight line braking. More toe-out will improve initial turn-in response but can overslip the inside tire during mid-corner resulting in excessive scrub. At the rear of the car only toe-in is available. More toe-in will result in greater stability during braking and on power movements however it may result in more mid-corner understeer and slower overall response to inputs.

